

Solar based on road **charging station.**



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Acknowledgement

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The internship gave me great exposure and look into the wide scope of hybrid and Electric vehicles. Finally I thank my parents for their constant support and sacrifices done to make me educated .

Declaration

I declare that this submission represents my own ideas in my own words. I have also included others ideas or words from their research papers. I have cited and referred their original sources at the end of my report. I have also adhered to principles of honesty and integrity. I have not misrepresented any data in my submission.

Eluru,
17-05-2021.

Kesava satish boppana

ABSTRACT

A solar powered charging station is to be designed so that devices can be charged outdoors and in an environmentally friendly way. This system converts solar energy to electricity and stores it in a battery bank.

A microcontroller prevents the batteries from being overcharged and prevents the system from being used when the batteries need charging. Electric vehicles (EVs) are becoming increasingly popular in many countries of the world. EVs are also proved to be more energy efficient and environmental friendly than ICVs.

As EV usage grows, more public spaces are installing EV charging stations. As solar has great potential to generate the electricity from PV panel, the charging of EVs from PV panels would be one of the solution and also a sustainable step toward the environment.

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Introduction:

In the past couple of years, advancements in technology put devices in our pockets that we could not have even dreamed of years ago. However, these devices often have drawbacks. One of the most pressing issues with phones, tablets, and laptop PCs is power.

We have not yet been able to develop efficient energy sources to match the efficiency of our devices. In fact, many laptops can drain a standard battery from a full charge in a matter of a couple hours. Here, I am proposing a solution that will provide power to charge devices in electric auto-mobiles using power generated from solar energy.

The paper presents charging standards including techniques and modes of solar charging, economic analysis, roadblocks and challenges, and sustainability, etc.

Purpose:

The objective of this project is to investigate the problem of providing an outdoor power source for charging devices in an environmentally friendly way to help reduce the demand of power from other methods.

From research of the existing designs, We found several points that can be improved. Three key areas that we saw were cost, efficiency, and complexity. Many will say the goal of electrical engineering is to design a cost effective, efficient, and easy to use device or product.

We saw this idea as an ingenious use of a product that will not only generate solar energy, but shade users from the harmful effects of sunlight. The only downfall is the panels are stationary and smaller, which may result in less energy produced.

Modelling:

It must have a source, a function, and an output. For our source, we will be using solar panels optimized with solar tracking. The system will contain the microcontroller to act as a charge controller and an inverter to convert from 12 Volt DC stored in the batteries to 110 Volt AC as the output.

The solar tracker would be affixed to the solar panel and would relay information to the microcontroller. The microcontroller would then send a signal to the stepper motor to adjust the angle of the solar panels. The microcontroller would also read the battery level and determine if the battery needs to be stopped from charging or stopped from being utilized, if it has been fully used.

A solar panel is a group of electrically connected photovoltaic cells made of semiconductor materials, such as silicon, which is currently most common. Multiple panels are electrically connected in either series to achieve a desired output voltage and/or in parallel to provide a desired current capability. A single solar cell produces only about 1/2 of a volt. A typical 12 volt panel about 25 inches by 54 inches will contain 36 cells wired in series to produce about 17 volts peak output.

EV and PHEV charging systems:

Infrastructure developments:

In the U.S., an official national goal of deploying **4 million** EVs on the road by **2023** has been established, and the government has implemented several policies to encourage public domain for achieving electrification at all levels.

Similarly, The German National Electric Mobility Platform (NPE) envisages around **2,000,000** EVs in Germany by **2024** with a demand of about 70,000 public on-street charging spots specifically.

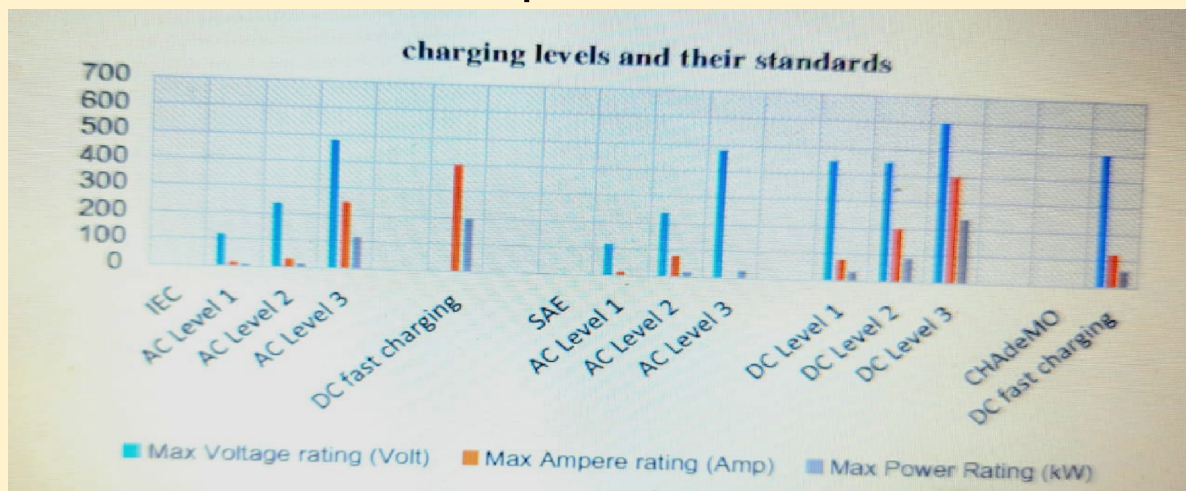
Also to way out the problems faced by the China's renewable energy exploitation practices and to cope with the growing energy demand of electric vehicles, model for setting up a number of solar powered charging stations for electric vehicles is designed.

In May 2017, 10 countries (Canada, China, France, Germany, Japan, Netherlands, Norway, Sweden, U.K, U.S.) have come together and formed a multi government policy forum i.e. Electric Vehicles Initiative (**EVI**), dedicated to accelerating the growth of electric vehicles worldwide.

Charging standards:

Plug-in hybrid electric vehicles (PHEVs) pave the way for partial electrification of the transportation sector. while, full electric vehicles (EVs) offers complete electrification of the transportation sector.

For plug-in electric vehicles (PEVs), on-board level-1 or level-2 chargers provide office daytime or home overnight charging, whereas high power off-board chargers provide fast charging. An on-board integrated charger, capable of fast charging for PEVs, can combine the advantages of both conventional on-board as well as off-board chargers. Focusing on the design of the vehicle, type and capacity of the battery used, different charging levels have been standardized by different organizations in the world. Accordingly, charging time and behavior varies from vehicle to vehicle. So, the design of PV-EV charger must be according to above-mentioned requirements.



Safety Aspects:

It is worthwhile to mention the fact that from distribution (consumer side) and grid (supply side) point of view the penetration of solar PV and EVs would further render the voltage fluctuations, frequency mismatching, and power quality issues, etc.

Typically, a battery charger has two stages: an **AC-DC** stage to rectify the grid AC voltage with power factor correction (PFC) and a **DC-DC** stage for regulating the battery currents and voltages. A **pwm DC/DC** converter for **6.6-kW EV Onboard Charger** is proposed in, the scheme has a similar structure as of conventional resonant converters, the only difference is that the controller uses PWM technique in place of a resonant controller.

The biggest constraint to this project will be to maximize the solar efficiency to provide the most power to the system that can be generated by the solar panels. Weather and solar patterns must be accounted for when making all of the calculations for the efficiency and output of the solar panels. **Climate factors**, such as clouds, moisture, haze, dust, and smog will have a degrading effect on the output power of the station's panel array.

Social and Economic aspects:

Obtaining the greatest amount of sunlight throughout the day needs to be for optimum output. Different enhancements to the solar panels such as adding solar concentrators or a solar tracking device may be necessary adding to the cost. Research on these devices is currently being done so that we may incorporate them into the final product while we test the smaller components of the charging station. We have to ensure the **efficiency** of the battery system which will be used to store the energy from the solar panels.

Ensuring proper safety regulations are met is another constraint dependent upon the electrical design, but more importantly is the overall structure of the station. Building and safety codes must be researched and implemented. A stand alone structure poses significantly less risk regarding fire safety when proper precautions are taken during site preparation. We have not yet conducted research into the structure of the system, but upon the successful testing of the electronics, this will be a major focus of the final product. Not only do we want a successful product, **but we want to make sure it is ethical and is safe for all.**

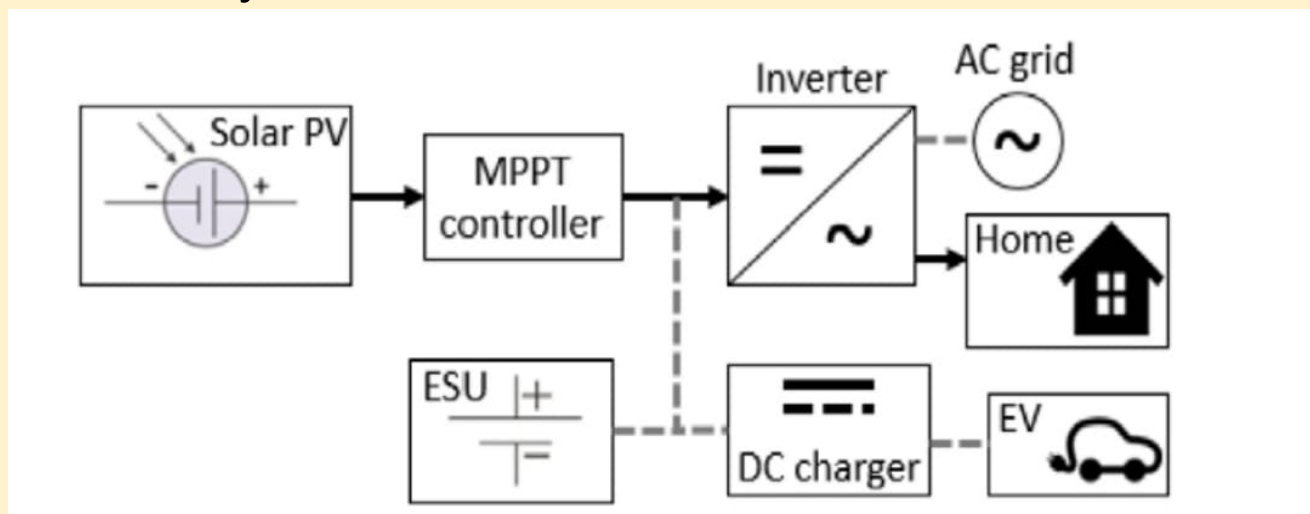
Available technologies:

Standalone-Backup Power PVCS:

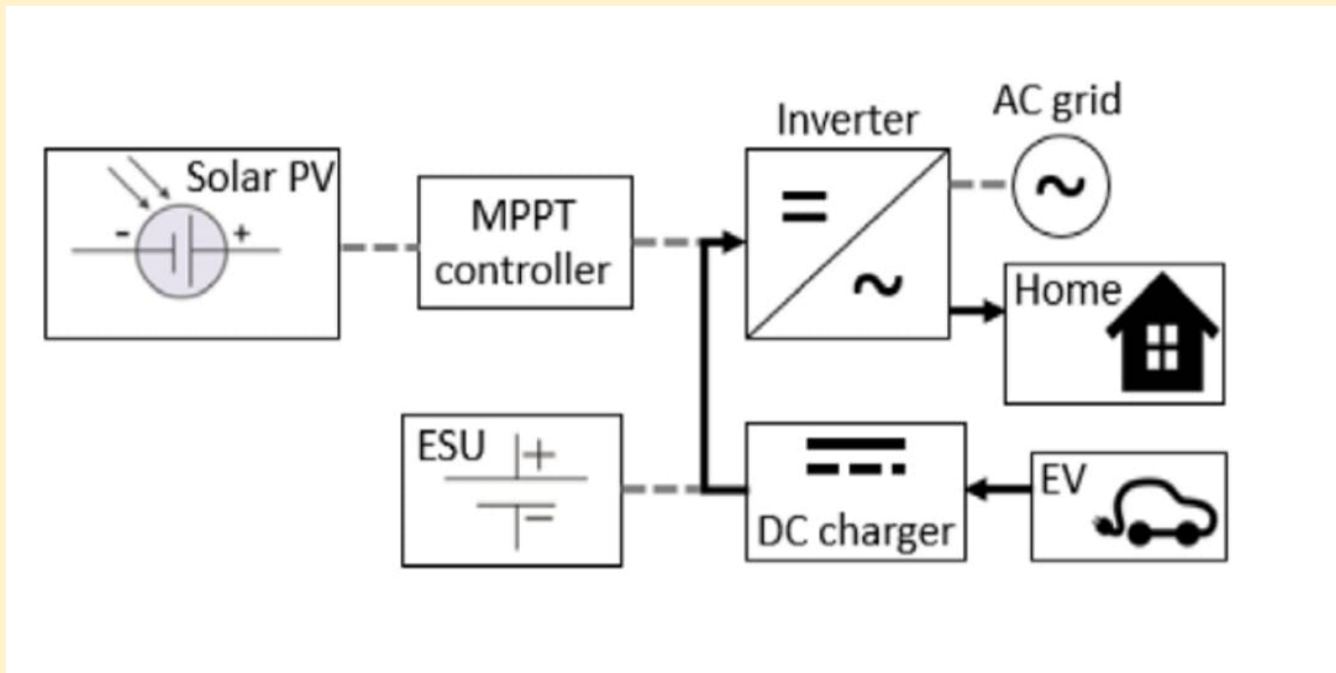
A stand-alone photovoltaic generation system is a complete set in which components are interconnected for directly converting solar irradiance into electricity. The components are PV generator, battery, charge controller, inverter, and the system load. The system is a PV power generation plant which is not connected to any utility grid.

Stand-alone PV system concepts found in the market are as follows: –

Solar Home System: Photovoltaic stand-alone, for V2H/H2V application, that can fully charge at home, the BEV overnight or can supply the residential load through EV battery.



All consumers and PV generator are coupled to the DC voltage – Small AC local grid with coupled components:



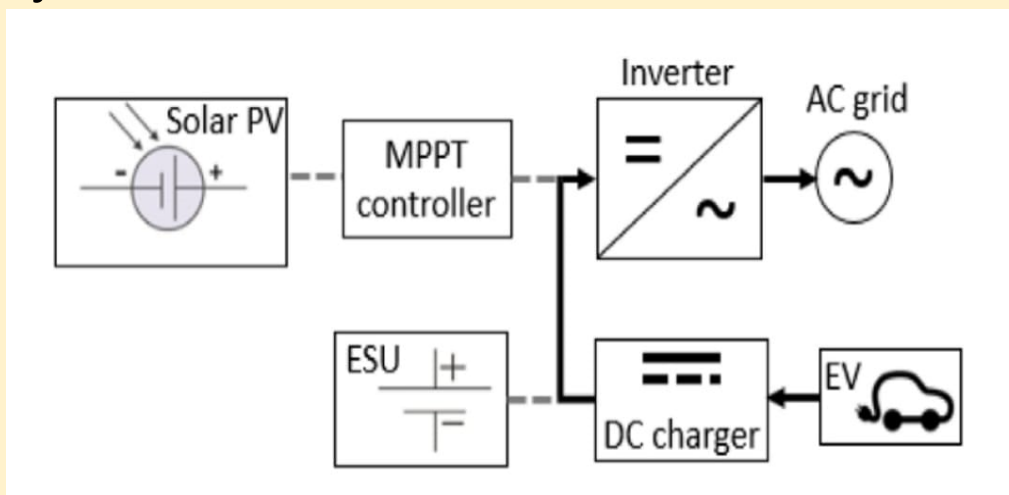
this PV technology appeared due to the desirable supply (minimum power) AC loads by DC power sources and to charge the battery on the DC from different sources.

Modular AC coupled systems:

more flexible PV systems with modularly structured components are achieved by coupling all consumers and a PV generator on the AC.

Solar Powered EV Charging System:

Electric vehicle (EV) charging from SPV offers a sustainable way for recharging the car batteries. Workplaces such as factories, office buildings, and industrial areas are perfect places to enable solar EV charging where the area under the building's rooftops and car parks can be utilized to install photovoltaic panels. The generated power is utilized directly for EV charging in an EV-PV charger, without the need for the storage system.



The literature on solar energy for charging the EVs is much advanced and diverse. This is because the electricity produced by solar PV provides more flexible integration with the existing grid. Among the various strategies, one is to charge the EV directly using the principle of 'charging-while-parking' to supplement the more common practice i.e. 'charging-by-stopping.'

Solar Powered DC Fast Charging System:

A bi-directional DC-DC electric vehicle (EV) charger is positioned between the high-voltage DC bus of a PV system and the EV battery. In addition, to charging the EV battery from the PV or the grid at a faster rate, the fast changes in PV power output to the battery can be diverted by the charger itself. Hence, the rate of change of inverter output power reduces to a level below the existing grid resource ramp rate. The EVs have inbuilt direct DC fast charging port and consequently, no inverter is needed between PV and EV. But in the on-grid system, the inverter is needed between PV and power grid.

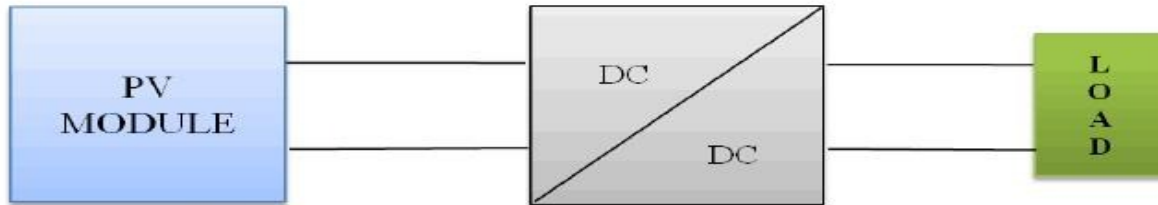


Fig. 1: Block diagram of Typical MPPT system

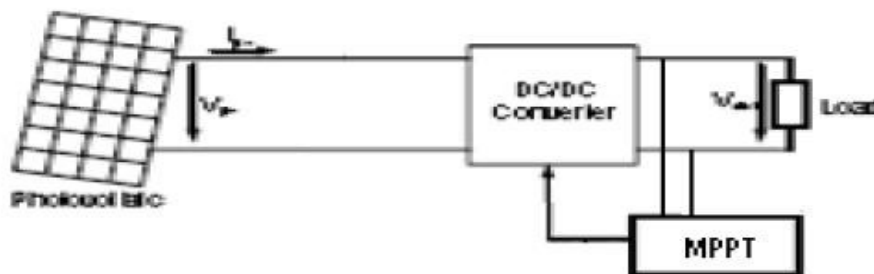
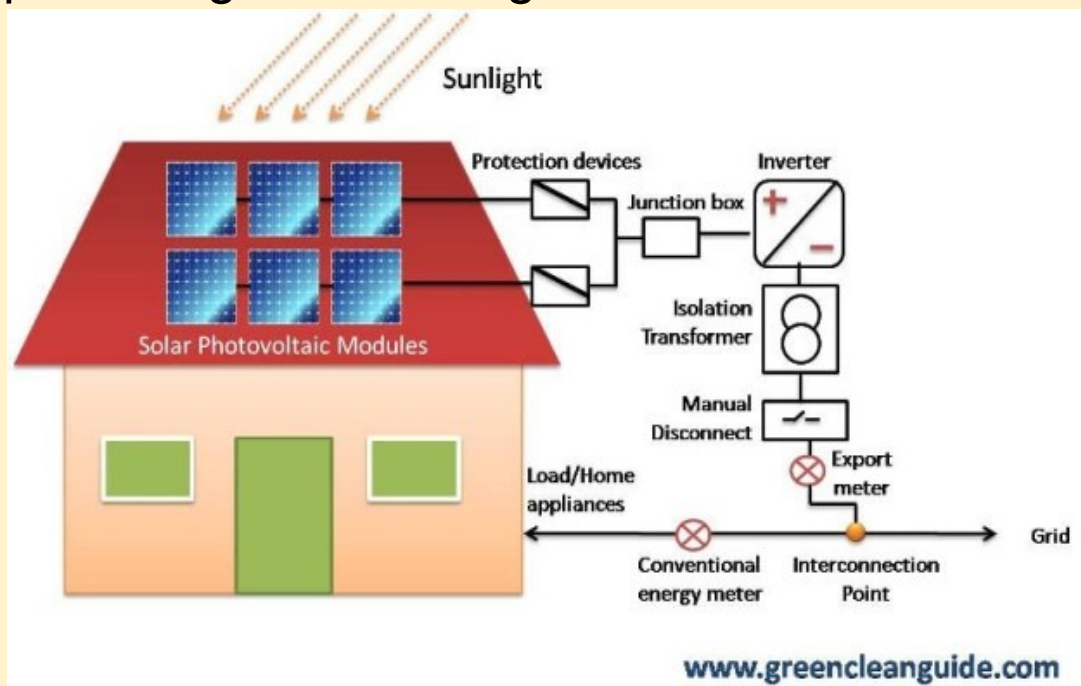


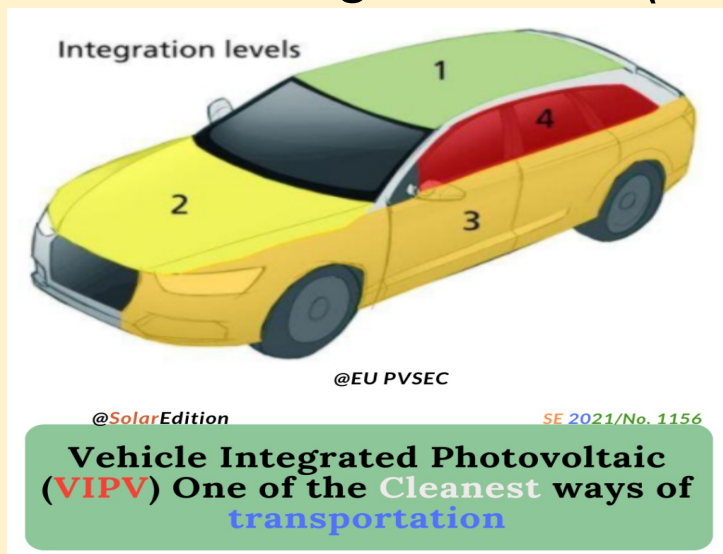
Fig. 2: DC – DC converter for operation at the MPPT

Solar Rooftop or Ground Mounted System:

It is the most widely used and practically accepted system in the world. In this system, the solar panels are installed on top of a roof (or) the land area can be utilized as a ground mounted system for the purpose of electrical or thermal energy generation. As far as the electrical generation is concerned, the system includes photovoltaic modules, mounting systems, cables, solar inverters, and other electrical accessories. The rooftop system is small, typically 5–20 kilowatts (kW) on residential buildings and 100 kilowatts(or more) mounted on commercial buildings while the ground-mounted system is sometimes very large ranging in megawatt capacities. A study for the installation of a 6KWp grid tie solar system capable of producing 25 units is given.



Vehicle-Integrated PV (ViPV) System:



The basic design of Vehicle-Integrated PV system defines the EV car's body as embedded with the solar photovoltaic material. Solar cells might be embedded into the chassis part exposed to sunlight like the hood, roof and maybe the trunk. "This would allow a hybrid vehicle to be partially powered using solar energy".

Hence, the vehicle itself resembles a small solar power plant and hybrid automobiles converted into even more environmentally friendly. Evidently, the sunnier the climate, the more the share of the contribution that solar cells might make fuel efficiency. Basically, it is a dynamic source which can supply additional power for electronic control units, actuators, displays, heating, ventilation, air conditioning units, and other electronic equipment such as 110 or 230VAC power converters, refrigerators and even microwaves, etc...

Solar Parking Lot:

Workplaces like offices, factories, industrial areas, and commercial buildings are perfect places to facilitate solar PV-EV charging where the building's rooftops and car-parking shed can be mounted with photovoltaic (PV) panels. Essentially a photovoltaic parking lot provides the on-site energy generation for supplying the electric car batteries and hence the power is directly available for EV charging through an EV-PV charger, without necessitating the energy storage.

Most of the vehicles are parked throughout the day at workplace parking spaces and these can be charged from the solar energy at normal charging rate using photovoltaic (PV)-based charging facility. Which can help to reduce the load and emissions from the power grid but also affects the cost of charging.



Some of the companies and MNCs/Government offices and tend to follow a 5-day, 8–9 h per day working schedule. which will be more than sufficient for an EV to get charged.

Economic and environmental effects:

Today, the world is facing two big transitions in generation and transportation systems simultaneously. The existing system is fossil fuel-based i.e. power generation from coal/nuclear etc. and transportation from oil and gas as propulsion power.

One of the main factors that propelled the increase in EVs' popularity is their contribution to reducing the greenhouse gas (GHG) emissions. HEVs and PHEVs have IC engines, their emissions are less than the conventional vehicles.

India accounts for more than 17% of the world's population, i.e. about 1.2 billion people. Hence, it is very challenging for India to provide sufficient energy supplies to all the consumers at an equitable cost. In India, about 70% of overall electricity generation is based on thermal energy. Huge amounts of non-renewable sources such as coal are being consumed for the production of thermal power which leads us to degradation of the environment.

Scenario generation using solar power is perceived to be too costly. Installation of solar equipment for the typical household is difficult as enough technical awareness is lacking in choosing suitable components with appropriate ratings.

From the perspective of the EV owners, EVs provide less operating cost because of their superior efficiency, it can be up to 70% where ICE vehicles have efficiencies in the range of 60–70%. The current high cost of EVs is likely to come down from mass production and better energy policies which will further increase the economic gains of the owners. V2G also allows the owners to obtain a financial benefit from their vehicles by providing service to the grid. The power service providers benefit from EV integration mainly by implementing coordinated charging and V2G. It allows them to adopt better peak shaving strategies as well as to integrate renewable sources. EV fleets can lead to \$200–\$300 savings in cost per vehicle per year.

Meanwhile, there is a major move toward large-scale decentralized renewable energy production through photovoltaic (PV) system due to dropping the cost of PV panels. The combination of EV and PV provides a unique opportunity for sustainable charging of electric vehicles. Many countries have set their benchmark as 2030–2050 and making their policies accordingly.

From an environmental viewpoint, renewable energy has a substantial amount of potential to decarbonize the NEA power system. Renewables' share grows from 20% in the Base case to 47%, resulting in the CO₂ emissions reductions from power generation of 36%.

Conclusion:

Solar PV and EVs are two sectors that turn out to be the future mode of transport and have a considerable potential for reducing CO₂ emissions and fossil fuel consumption. For PV, its variable nature and the depth of electricity output in the daytime restricts its contribution in meeting a large fraction of typical electricity demand.

Though EVs provide a clean, efficient, and noise free means for commuting for when compared with the gasoline vehicle. This paper has presented a review of solar powered EV systems, various aspects such as charging infrastructure and EV energy management has been discussed in detail. The key challenges include the cost and performance of the battery storage systems that restrict their economic consumption. And also has a limited utility as well owing to driver range anxiety and charging infrastructure access.

The driver range anxiety significantly affects the selected utility and may vary subject to the average high and low range anxiety drivers. Across these scenarios, the potential gains of different solar PV-EV charging techs have highlighted, yet not cater the vehicle utility and infrastructure cost relationship for affordable, more realistic models setup. Hence, great future prospects are there regarding these mentioned roadblocks to mitigate out.

Social barriers also check the wide adoption of this technology, particularly in developing countries. People have a tendency to resist new technologies that are considered unaware/unaccustomed, consequently, the effective way will be the policy-making decisions that show their critical concerns.

There are still a number of technical problems to overcome, particularly in the area of power electronics i.e. bidirectional converters and motor drive propulsion systems in EVs and PHEVs. Despite these challenges, if both these technologies deployed together, the improved economic performance of EVs might be possible by the availability of daytime charging, increasing the distance travelled using low-cost electricity and potentially reducing the size of the battery.

Accelerating the deployment of renewable energy and the electric vehicle will fuel economic growth, create new employment opportunities, enhance human welfare, and contribute to a climate-safe future.

Nomenclature:

- **H.E.V = hybrid and electric vehicles**
- **E.V = electric vehicles**
- **P.V = photovoltaics**
- **P.H.E.V = plug in hybrid electric vehicle**

- **I.C.E** = internal combustion engine
- **N.P.E** = nationale programme of e.v's
- **CO2** =carbon di-oxide
- **G.H.G** =greenhouse gases
- **S.P.V** =solar powered vehicles
- **B.E.V** =battery electric vehicles
- **M.P.P.T** =maximum power point tracking
- **E.S.U** =electrostatic units
- **V2G** =vehicle to grid technology
- **P.V.C.S** = photovoltaic current source

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Know your EV Charging Stations

 AC Level 1	 AC Level 2	 DC Fast Charge
Voltage 120V 1-Phase AC	Voltage 208V or 240V 1-Phase AC	Voltage 208V or 480V 3-Phase AC
Amps 12 – 16 Amps	Amps 12 – 80 Amps (Typ. 32 Amps)	Amps <125 Amps (Typ. 60 Amps)
Charging Loads 1.4 to 1.9 kW	Charging Loads 2.5 to 19.2 kW (Typ. 7kW)	Charging Loads <90 kW (Typ. 50kW)
Charge time for vehicle 3 – 5 miles of range per hour	Charge time for vehicle 10 – 20 miles of Range per hour	Charge time for vehicle 80% Charge in 20 – 30 minutes

*Thank you....
Kesava satish boppana*